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Taylor Approximation of Inventory Policies for One-Warehouse, Multi-Retailer Systems with Demand Feature Information

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Abstract. We consider a distribution system in which retailers replenish perishable goods from a warehouse, which, in turn, replenishes from an outside source. Demand at each retailer depends on exogenous features and a random shock, and unfulfilled demand is lost. The objective is to obtain a data-driven replenishment and allocation policy that minimizes the average inventory cost per time period. The extant data-driven methods either cannot guarantee a feasible solution for out-of-sample feature observations or generate one with excessive computational time. We propose a policy that resolves these issues in two steps. In the first step, we assume that the distributions of features and random shocks are known. We develop an effective heuristic policy by using Taylor expansion to approximate the retailer's inventory cost. The resulting solution is closed-form, referred to as *Taylor Approximation* (TA) policy. We show that the TA policy is asymptotically optimal in the number of retailers. In the second step, we apply the linear quantile regression and kernel density estimation to the TA solution to obtain the data-driven policy called *Data-Driven Taylor Approximation* (DDTA) policy. We prove that the DDTA policy is consistent with the TA policy. A numerical study shows that the DDTA policy is very effective. Using a real data set provided by Fresh Hema, we show that the DDTA policy reduces the average cost by 11.0% compared with Hema's policy. Finally, we show that the main results still hold in the cases of correlated demand features, positive lead times, and censored demand.

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Keywords: distribution system • Taylor approximation • data-driven • inventory allocation

1. Introduction

Fresh Hema, a subsidiary of Alibaba Group, is a grocery chain that owns more than 200 stores and offers fresh goods to customers in more than 20 major cities in China as of 2021. The focal logistics system that motivates this research is a central warehouse located in Chengdu city which is responsible for inventory replenishment for 16 retail stores scattered in the city. The company adopts a centralized inventory system in which the warehouse orders fresh goods, such as vegetables, eggs, and meat, from local farmers every morning. The ordered products usually arrive at the warehouse in the early morning of the following day and then are immediately loaded onto trucks that deliver the products to the retail store by 6:00 a.m. on the same day. The fresh goods market is very competitive in China because there are many traditional, local fresh goods markets in high-population areas. Fresh goods usually have very short shelf lives; therefore, the company requires an accurate demand estimation for inventory replenishment to stay competitive. For this reason, the company has started to collect feature information for retail stores such as past sales, demographics, and page views in recent years. The feature information plays a key role in the inventory decision as it can help predict the next day's demand for the retail store. For example, when the company observes a larger volume of page views for an item, the demand for the item for the following day is often higher. Nevertheless, the company is seeking a simple and implementable inventory policy that utilizes the feature data. This is the question that we aim to address in this paper.

There is a long history in the inventory literature of the distribution system without demand feature

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information. It is known that the optimal policy for the general lead-time model with backorders, if it existed, would be very difficult to characterize. Nevertheless, Clark and Scarf (1960) show that, under the so-called balance assumption, the myopic allocation is optimal and the optimal policy can be obtained as follows. First, each retailer computes its optimal base-stock level as if it has ample supply from the warehouse. Then, an induced penalty cost is charged to the warehouse for not being able to allocate enough inventory to reach the optimal base-stock level of the retailer. Finally, the warehouse solves a constrained optimization problem by minimizing the sum of its holding cost and the induced penalty costs to determine the optimal order quantity and allocation policy. If our model does not consider the feature information, the model satisfies the balance assumption because excess inventory is disposed of and unfilled demand is lost. The optimal policy turns out to be quite simple: the warehouse orders an amount equal to the sum of the optimal order quantities of retailers and ships each retailer the ordered quantity after it arrives. (See the *No-Feature* (NF) model in Section 3.) That is, there is no allocation decision to make, and the warehouse becomes a cross-docking location. In fact, Fresh Hema adopts this idea to implement its inventory policy. Specifically, Hema uses machine learning methods to estimate the demand function of features based on historical data. At the beginning of the day, the retailer observes the features, calculates the order quantity based on the demand function according to the newsvendor solution, and places an order with the warehouse, which ships the ordered inventory to the retailer the following day. In other words, Hema adopts cross-docking and uses *past* feature information for inventory replenishment. Although Hema's approach is easy to use, one natural question is whether there exists a data-driven policy that outperforms Hema's current practice. Intuitively, the warehouse can better allocate inventory available according to more accurate demand information after the *current* features are observed. As shown in this paper, the allocation decision plays a critical role in the current model when the feature information is present.

We consider a one-warehouse, multiretailer system in which a central warehouse orders from an outside unlimited source and allocates inventory to J nonidentical retailers which face random demands. There is one period of lead time for the warehouse and zero lead time for the retailers, that is, same-day delivery. (A general lead-time case is discussed in Section 6.) Demand for each retailer j is correlated with exogenous features and a random shock. Unfulfilled demand is lost and excess inventory is disposed of. There are linear holding costs and lost-sales costs evaluated at the end of a period. An inventory planner places an order for the warehouse at the beginning of a period. The order arrives the following day, and an allocation decision is made after observing the random demand features. The objective is to obtain a data-driven, feature-dependent replenishment and allocation policy such that the long-run average cost per period is minimized.

Using the extant data-driven methods in our model to provide the inventory solution (prescriptive predictor) for given features encounters challenges. Specifically, one may use the empirical risk minimization (ERM) approach (e.g., Ban and Rudin 2019) to train the inventory solution by solving the simple average cost of N-constrained optimization problems where N is the in-sample data size. However, the computational time is extensive if J and N are large. In addition, the resulting solution may not satisfy the allocation constraint for any out-of-sample feature observations. Alternatively, one may adopt the local learning (LL) framework of Bertsimas and Kallus (2020), which uses in-sample data to train the weight functions for given features. Although this approach can generate a feasible solution, like ERM, the LL approach also needs to solve N-constrained optimization problems in the training step, which incurs significant computational time. (Please refer to Section 7 for a numerical study.)

We propose a two-step procedure to tackle these challenges and generate a simple and implementable data-driven policy. The key idea is to integrate the allocation decision into the inventory solution through Taylor expansion. Specifically, in the first step, we assume distributions are known and approximate the retailers' cost functions by Taylor expansion. Solving the problem with the approximated cost functions, we can derive a closed-form solution for the order and allocation decisions for the warehouse. We refer to the resulting heuristic as the *Taylor Approximation* (TA) policy. The required components to execute the TA policy are the retailers' base-stock levels and the density for given features at the base-stock level. Thus, in the second step, we apply machine learning methods to estimate these required components. Specifically, we use linear quantile regression to generate the retailers' base-stock levels and kernel density estimation to generate the density. We refer to the resulting data-driven policy as the *Data-Driven Taylor Approximation* (DDTA) policy.

This work made the following contributions. First, with the known demand distributions, we show that the TA policy is asymptotically optimal in the number of retailers (Theorem 1). This result is established by showing that the difference between the total cost of the TA policy and that of a *lower-bound* system is bounded by a constant. The lower-bound system is constructed by assuming that the demand feature information is known in advance when the warehouse replenishes inventory at the beginning of a period. (See the *Advanced Feature* (AF) model in Section 3.)

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With this assumption, the warehouse knows precisely the retailers' order quantities generated by the assumed known feature information so that the resulting cost is a lower bound to the optimal cost.

Second, we extend the data-driven policy with feature information from single-stage systems to distribution systems. The DDTA policy originates from the TA policy that bypasses extensive computations on the inventory allocation decision. We show the convergence rate for the estimation error on the total cost of the DDTA policy to that of the TA policy (Theorem 2). We also derive the generalization bound, that is, the difference between the generalization error (or the out-of-sample cost) and the empirical in-sample cost, for the DDTA policy (Theorem 3). Combining Theorems 1–3, we show the effectiveness of our simple data-driven policy. We believe that such a two-step framework, that is, obtaining a simple solution to ensure feasibility and effectiveness and applying machine learning methods to the simple solution, can be applied to other complex constrained optimization problems.

Finally, we conduct a case study by applying our DDTA policy to a real data set provided by Fresh Hema. We compare our policy with Fresh Hema's current policy and find that the DDTA policy can generate a solution in less than a second and outperform Hema's policy by 11.0%.

The rest of the paper is organized as follows. Section 2 reviews the literature. Section 3 introduces the model. Section 4 derives the TA policy when the demand and feature distributions are known. Section 5 applies data-driven methods based on the TA policy to generate the DDTA policy. Section 6 considers several extensions by incorporating correlated demand features, general lead times, and censored demands. Section 7 numerically tests the effectiveness of the DDTA policy compared with other data-driven policies. Section 8 conducts a case study based on Fresh Hema's data. Section 9 concludes the paper. Online Appendix A presents the proofs for our main results. Online Appendix B provides discussions, proofs, and the details omitted in the paper.

2. Literature Review

We study a distribution system with nonidentical retailers whose demands are correlated with exogenous features. There are two closely related streams of literature: distribution systems with known demand distribution and data-driven inventory models.

The distribution inventory system with known demand distribution has been studied extensively since the seminal work of Clark and Scarf (1960). It is well known that, under the so-called balance assumption, that is, that inventory among retailers can be transferred instantaneously and costlessly, the optimal joint replenishment and allocation policy is that the warehouse orders up to an echelon base-stock level and allocates its inventory according to the myopic allocation rule. Nevertheless, without the balance assumption, the optimal policy remains unknown. Thus, there is a stream of research aiming to propose simple and effective heuristic policies or derive lower bounds to the optimal cost, for example, Nahmias and Schmidt (1984), Jackson (1988), Chen and Zheng (1994), Axsäter et al. (2002), and Gallego et al. (2007). A paper more relevant to our model is Marklund and Rosling (2012), who consider a one-warehouse–multiretailer system in a finite horizon. They propose an approximate model by revising the allocation constraint in the Clark-Scarf model. Specifically, the authors constrain the sum of expected inventory positions and show that the resulting cost is a lower bound to the optimal one. There are many variants to the classic Clark-Scarf model. For example, Perakis et al. (2020) consider inventory allocation of multiple products across a network of warehouses with both the inventory depth and capacity constraints. They propose an iterative algorithm that converges to the optimal solution. Govindarajan et al. (2021) consider a multilocation newsvendor network when only the mean vector and covariance of the joint distribution of demands are known. They adopt a distributionally robust model to find inventory levels that minimize the worst-case expected cost. Zhu et al. (2021) evaluate a class of easy-to-implement heuristics, called modified echelon (r, Q) policies, and show that the proposed policy is asymptotically optimal, as pairs of system primitives, such as the ratios of the fixed cost of the central facility to those of the local facilities, are scaled up. Miao et al. (2022) consider an allocation problem for a fixed amount of inventory in a finite-time horizon. They show that the allocation policy based on Lagrangian relaxation is asymptotically optimal when the numbers of retailers and time periods approach infinity. Nambiar et al. (2021) study a similar problem and propose a computationally tractable algorithm under demand learning by using a Lagrangian relaxation approach. We refer the reader to Simchi-Levi and Zhao (2012) for a survey on the classic multiechelon inventory models. Our contribution to this stream of literature is that we propose a closed-form replenishment and allocation policy based on Taylor expansion, and show that the heuristic is asymptotically optimal. Note that Marklund and Rosling (2012) and Miao et al. (2022) have derived similar asymptotic results with different approaches, but they do not consider demand with feature information.

The literature on data-driven applications for inventory problems mainly focuses on newsvendor or single-stage models. This literature can be further categorized into whether there are feature data for the demand model. For the model without feature information,

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Liyanage and Shanthikumar (2005) consider a newsvendor model. Assuming the sample data are generated from a family of demand distributions, they derive the optimal order function for given data observations and optimize the parameters of the order function to derive the so-called "operational statistics" as the data-driven inventory solution. Levi et al. (2007) obtain the data-driven newsvendor solution based on sample average approximation (SAA). They derive the performance bound with respect to the sample size, and later Levi et al. (2015) improve the finite sample bound. Huh et al. (2011) consider a newsvendor model with censored demands. They use the Kaplan-Meier estimator to derive an adaptive data-driven policy and show that the policy converges to the optimal one with the known demand distribution. Zhang et al. (2019) consider the same model of Huh et al. (2011) and propose a new learning algorithm that has a square-root convergence rate that matches the theoretical bound. Chen and Chao (2020) study the optimal learning algorithms under substitution for multiple products with censored demand. This stream of research recently has been extended to nonstationary systems. Keskin et al. (2021) study a newsvendor model with unknown, time-varying demands over a finite horizon. They develop a nonparametric dynamic ordering policy with a time-moving window, which tracks the changes in the demand rate over time.

For the model with demand feature information, Ban and Rudin (2019) consider a newsvendor model and assume that the optimal replenishment policy is a linear function of features. They apply ERM and kernel-weight optimization to generate the order quantity and show the value of feature information as well as the performance bounds on the out-of-sample costs. Bertsimas and Kallus (2020) propose a general framework for using data to prescribe optimal decisions. Their idea is to use nonparametric machine learning algorithms, such as k-nearest neighbor (kNN) and regression trees, to find the weight function based on features and use it to solve the weighted average of the objective value for out-of-sample data. They show the data-driven solution is asymptotically optimal when the sample size is large. Oroojlooyjadid et al. (2020) combine the ideas in deep learning and use the deep neural network to approximate decision function. Qi et al. (2020) extend the model in Ban and Rudin (2019) by considering nonstationary and intertemporal demand and feature data with an unbounded domain. Bertsimas and Koduri (2021) further propose two methods based on regression in reproducing kernel Hilbert spaces for solving the optimization problem with feature data. They refer to their methods as "global" machine learning methods as opposed to "local" ones in Bertsimas and Kallus (2020), which fit a functional form of the prediction model rather than approximate objective functions based on local points near the observations. They provide theoretical guarantees for both methods under general demand functions. Elmachtoub and Grigas (2022) provide a "smart" predict-then-optimize (SPO) framework by introducing the SPO loss function which measures the cost of predicting error in the decision for an optimization problem. Kannan et al. (2022) study the data-driven SAA problem with covariates by the "predict-then-smart-optimize" (PTSO) framework.

Another stream of data-driven inventory research adopts a distributionally robust optimization framework (DD-DRO) and minimizes the worst-case expected cost over some specifically chosen distribution ambiguity set. We refer to Rahimian and Mehrotra (2019) for a detailed discussion. Qi et al. (2022) focus on a similar model with the "fixed design" features instead of the "random design." Zhang et al. (2023) study a distributionally robust feature-based newsvendor problem and provide a class of optimal and interpretable policies with finite sample guarantees.

The data-driven research for the multilocation inventory problem is relatively sparse. Acimovic and Graves (2015) study a fulfillment problem and develop a heuristic that minimizes the immediate outbound shipping cost and an estimate of future expected outbound shipping costs. They use a company's data to estimate the required parameters in their heuristic. The general framework in Bertsimas and Kallus (2020) can be used to solve the inventory allocation problem for the multi-retailer model with demand features. However, as stated earlier, their algorithm requires extensive computations because of solving the allocation problem for each in-sample data observation during training. Notz and Pibernik (2022) study a capacity planning and allocation problem in a finite horizon. They formulate the problem into a two-stage optimization problem with recourse and propose two machine learning algorithms with out-of-sample guarantees based on sample average approximation and kernelized empirical risk minimization. However, their objective function is quite different from ours. Our contribution to this stream of research is that we extend the data-driven inventory policy to the distribution system with demand features.

3. Model and Problem Formulation

We consider a firm that manages its inventory system composed of one central warehouse and J nonidentical retailers over T periods. For notational convenience, we use $[J]^+$ and $[J]$ to denote the sets of $\{1,2,\ldots,J\}$ and $\{0,1,\ldots,J\}$, respectively, and $j \in [J]$ to denote a specific location, where location $j \in [J]^+$ is retailer j and $j = 0$ is the warehouse. Retailers replenish inventory from the warehouse, which, in turn, replenishes from an outside supplier with ample stock. The replenishment lead time for retailer j is zero (overnight delivery), and for

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the warehouse is one period. Retailers face random demands which depend on observable and exogenous random features, such as weather, temperature, and page views.[1] We use $x_j \in \mathbb{R}^{P_j}$ to denote retailer j's demand feature vector, where P_j represents the dimension of feature vector and $[P_j]^+ \triangleq \{1,2,\ldots,P_j\}$. Demand features are independent between retailers and i.i.d. between time periods. Unmet demands at retailers are lost and unsold inventories are disposed of at the end of each period. For retailer j, there is a holding cost h'_j for each unit of on-hand inventory and a lost-sales cost b_j for each unit of lost sales in each period. For the warehouse, holding cost h_0 is incurred for each unit of on-hand inventory in each period. Without loss of generality, we assume that the variable ordering cost is zero.

We use $t = 1,2,\ldots$ to denote the time index. Recall that the unsold inventory does not carry over to the next period and the lead time of the warehouse is one period. For the lost-sales system, the long-run average total cost per period is equal to the expected cost of a regenerative cycle that starts from the warehouse's order decision. More specifically, at the beginning of period t, the warehouse places an order z_0^t. This order will be received at the beginning of period $t+1$. Then, the central planner observes feature vector x_j^{t+1}, which affects the demand in period $t+1$, and makes the allocation decisions $z_j^{t+1}, j \in [J]^+$. The holding cost rate h_0 is incurred for the excess inventory in the warehouse after allocation. The allocated inventory will be dispatched and received at retailer j at the beginning of period $t+1$ (same-day delivery). Then, the demand $D_j^{t+1}(x_j^{t+1})$ realizes and inventory holding cost h'_j is incurred for each unit of unsold inventory and lost-sales cost b_j is incurred for each unit of lost sales at the end of period $t+1$. We assume that the cost parameters satisfy the conditions that $0 \le h_0 \le \underline{h} \le h'_j \le \overline{h} \le b_j \le \overline{b} < \infty$ for all $j \in [J]^+$, where $\underline{h} \triangleq \min_j h'_j$, $\overline{h} \triangleq \max_j h'_j$, and $\overline{b} \triangleq \max_j b_j$. We aim to determine the optimal z_0^t and $z_j^{t+1}, j \in [J]^+$ that minimize the expected total cost per period.

Consider the demand model: for any period t, let $D_j^t(x_j^t)$ denote retailer j's random demand, where

$$D_j^t(x_j^t) = \gamma_j^t + g_j(x_j^t) + \epsilon_j^t. \tag{1}$$

Here, γ_j^t is a constant, $g_j(\cdot)$ is a function of features, and ϵ_j^t is the random noise with zero mean i.i.d. over period t and independent over retailer j.

Consider a regenerative cycle starting from period t. Given realized feature vector x_j^{t+1}, the expected inventory costs at period $t+1$ can be expressed as follows:

$$h_0 \cdot \left(z_0^t - \sum_{j=1}^J z_j^{t+1} \right) + \sum_{j=1}^J \mathbb{E}_{\epsilon_j^{t+1}} [h'_j \cdot (z_j^{t+1} - D_j^{t+1}(x_j^{t+1}))^+ + b_j \cdot (D_j^{t+1}(x_j^{t+1}) - z_j^{t+1})^+ | x_j^{t+1}], \tag{2}$$

where $z_0^t \ge \sum_{j=1}^J z_j^{t+1} \ge 0$. The first term represents the holding cost at the warehouse, and the second term represents the expected holding and lost-sales costs at retailers.

For notational convenience, we simplify $D_j^{t+1}(x_j^{t+1})$ as D_j^{t+1}. The expected costs in (2) can be expressed in a more compact echelon form,

$$h_0 \cdot z_0^t + \sum_{j=1}^J C_j(z_j^{t+1} | x_j^{t+1}), \tag{3}$$

where

$$C_j(z_j^{t+1} | x_j^{t+1}) \triangleq \mathbb{E}_{\epsilon_j^{t+1}} [-h_0 \cdot D_j^{t+1} + h_j \cdot (z_j^{t+1} - D_j^{t+1}) + (b_j + h'_j) \cdot (D_j^{t+1} - z_j^{t+1})^+ | x_j^{t+1}], \tag{4}$$

and $h_j \triangleq h'_j - h_0$. According to our demand model in (1), the term $\mathbb{E}_{\epsilon_j^{t+1}}[-h_0 \cdot D_j^{t+1} | x_j^{t+1}] = -h_0 \cdot (\gamma_j^t + g_j(x_j^{t+1}))$ is a constant given the observed feature vector x_j^{t+1} and will not affect our decisions. Thus, we omit this term in the subsequent analysis.

We can formulate the one-cycle problem as two-stage stochastic programming. Specifically, for each cycle t, the warehouse first determines the replenishment quantity z_0^t and then the allocation decision z_j^{t+1} after observing realized demand features $x^{t+1} \triangleq (x_1^{t+1}, x_2^{t+1}, \ldots, x_J^{t+1})$. The first- and second-stage optimization problems are respectively given by

$$\textbf{Stage 1:} \quad \min_{z_0^t > 0} \; h_0 \cdot z_0^t + \mathbb{E}_{x^{t+1}} [G(z_0^t | x^{t+1})], \tag{5}$$

$$\textbf{Stage 2:} \quad G(z_0^t | x^{t+1}) \triangleq \min \sum_{j=1}^J C_j(z_j^{t+1} | x_j^{t+1}) \quad \text{s.t.} \quad \sum_{j=1}^J z_j^{t+1} \le z_0^t; \quad z_j^{t+1} \ge 0, \text{ for any } j \in [J]^+. \tag{6}$$

The expectation over x^{t+1} in (5) is because the inventory manager has not observed the feature information at the beginning of period t, and we consider the expected cost per period among all possible feature combinations.

We now introduce two additional systems which are relevant to our subsequent discussions: the advanced feature model and the no-feature model.

3.1. AF Model

In the advanced feature model, we assume that the feature information in period $t+1$ is known by the manager at the beginning of period t before the warehouse makes the inventory decision. That is, unlike the setting in which the warehouse's order decision is based on

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the expectation over $\boldsymbol{x}^{t+1}$, the warehouse can make an order directly based on the realization of $\boldsymbol{x}^{t+1}$. Consequently, the objective function of the AF model is (3) and (4) for any given $\boldsymbol{x}^{t+1}$. We can show that the cost under the AF model is a lower bound to that of our original system. Further, through the Karush-Kuhn-Tucker (KKT) conditions, the optimal policy is that the retailer orders according to a newsvendor solution, that is,

$$s_j^l(\boldsymbol{x}_j^{t+1}) = F_j^{-1}\left(\frac{b_j}{b_j+h_j'}\middle|\boldsymbol{x}_j^{t+1}\right), \tag{7}$$

where $F_j(\cdot|\boldsymbol{x}_j^{t+1})$ is the conditional cumulative distribution function (CDF) of demand D_j. We call $s_j^l(\boldsymbol{x}_j^{t+1})$ the *local* base-stock level.[2] That is, there is no allocation decision to make and the warehouse serves as a cross-docking point. We refer the reader to the proof of Theorem 1 for a detailed derivation.

3.2. NF Model

Suppose that we do not have the feature information, and each retailer faces a random demand D_j^t in period t. Consider a regenerative cycle starting from period t: the objective function is

$$h_0 \cdot z_0^t + \sum_{j=1}^{J} C_j(z_j^{t+1}), \tag{8}$$

where

$$C_j(z_j^{t+1}) = \mathbb{E}_{D_j^{t+1}}[h_j \cdot (z_j^{t+1} - D_j^{t+1}) + (b_j + h_j') \cdot (D_j^{t+1} - z_j^{t+1})^+]. \tag{9}$$

This is a special Clark-Scarf model with the warehouse's lead time equal to one period and perishable goods/lost sales. Similarly, one can show that through the KKT conditions, the structure of the optimal policy is the same as that of the AF model. Clearly, the demand variability is larger in the NF model as no feature information is available. We omit the proof for brevity.

Our goal is to derive an easy-to-implement data-driven policy for distribution systems with only demand and feature data available. To achieve this, we first explore the structure of optimal policy for the system assuming that the demand and feature distributions are known. Because of the random demand features, the optimal policy becomes computationally intractable, and thus we propose a simple heuristic named the *Taylor Approximation* policy in Section 4. We show that the TA policy is asymptotically optimal in the number of retailers. Based on the TA policy, we propose a data-driven predictor named the *Data-Driven Taylor Approximation* policy in Section 5. We evaluate the performance of the DDTA policy by comparing the difference between the expected optimal cost denoted by $\mathbf{C_{OPT}}$ for the system with known distributions and the empirical in-sample cost[3] under the DDTA policy denoted by $\mathbf{R_{DDTA}}(S_N)$, where S_N is the training data set. Equation (10) illustrates the roadmap of the analysis for the performance bound $|\mathbf{R_{DDTA}}(S_N) - \mathbf{C_{OPT}}|$. To analyze the performance, we introduce the expected inventory cost under the TA policy, denoted by $\mathbf{C_{TA}}$, for the system with known distributions and the out-of-sample inventory cost under the DDTA policy, denoted by $\mathbf{C_{DDTA}}(S_N)$, which is also named as the generalization cost (error) in the machine learning literature; see, for example, Bousquet and Elisseeff (2002). The performance bound consists of three components: the optimality gap (Theorem 1 in Section 4), which represents the performance of the TA policy compared with the optimal cost; the estimation error (Theorem 2 in Section 5), which measures the expected cost of the DDTA policy compared with that of the TA policy;[4] and the generalization bound (Theorem 3 in Section 5), which measures the gap between the empirical cost and the generalization cost for the DDTA policy.

$$\begin{aligned}&|\mathbf{R_{DDTA}}(S_N) - \mathbf{C_{OPT}}|\\ &\le \underbrace{|\mathbf{C_{TA}} - \mathbf{C_{OPT}}|}_{\text{Optimality Gap}} + \underbrace{|\mathbf{C_{DDTA}}(S_N) - \mathbf{C_{TA}}|}_{\text{Estimation Error}}\\ &\quad + \underbrace{|\mathbf{R_{DDTA}}(S_N) - \mathbf{C_{DDTA}}(S_N)|}_{\text{Generalization Bound}}.\end{aligned} \tag{10}$$

4. The Model with Known Distributions

In Section 4.1, we study the optimal solution for the two-stage stochastic optimization problem in (5) and (6) when the demand and feature distributions are known. Because of the computational complexity, in Section 4.2, we then propose a simple heuristic policy, the Taylor approximation policy. Section 4.3 derives the optimality gap $|\mathbf{C_{TA}} - \mathbf{C_{OPT}}|$ between the optimal cost $\mathbf{C_{OPT}}$ and the expected cost $\mathbf{C_{TA}}$ under the TA policy. We omit the superscript t later for simplicity.

4.1. Optimality Analysis

We first solve the second-stage problem in (6) given z_0 and the feature vector $\boldsymbol{x}_j$. It is well known that the objective function in (6) is convex in z_j and the optimal solution satisfies the KKT conditions. Define $s_j^e(\boldsymbol{x}_j) \triangleq F_j^{-1}\left(\frac{b_j+h_0}{b_j+h'_j}\middle|\boldsymbol{x}_j\right)$ as the echelon base-stock level for retailer j, which depends on demand feature $\boldsymbol{x}_j$, and we abbreviate it as s_j^e. Without loss of generality, we assume that $b_1 \le b_2 \le \cdots \le b_J$. The following proposition characterizes the structure of the optimal allocation policy.

Proposition 1 (Optimal Allocation Policy). *For any $k \in [J]^+$, define $a_k \triangleq \sum_{i=k}^{J} F_i^{-1}\left(\frac{b_i-b_k}{b_i+h'_i}\middle|\boldsymbol{x}_i\right)$, and by definition $a_1 \ge a_2 \ge \cdots \ge a_J = 0$. Given the warehouse's inventory*

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level $z_0 > 0$ and realized feature $\boldsymbol{x}$, the optimal allocation policy $z_j^*(z_0, \boldsymbol{x})$ for any retailer $j \in [J]^+$ and the optimal Lagrange multiplier $\lambda^*(\boldsymbol{x})$ are shown as follows:

- *When $z_0 > a_1$: $0 \le \lambda^*(\boldsymbol{x}) < (b_1 + h_0)$, and*

$$z_j^*(z_0, \boldsymbol{x}) = \begin{cases} F_j^{-1}\left(\dfrac{b_j + h_0 - \lambda^*(\boldsymbol{x})}{b_j + h_j'}\middle|\boldsymbol{x}_j\right), & \text{if } z_0 < \sum_{j=1}^J s_j^e; \\ s_j^e, & \text{if } z_0 \ge \sum_{j=1}^J s_j^e. \end{cases} \tag{11}$$

- *When $a_{k+1} < z_0 \le a_k$ for $k \in [J-1]^+$: $(b_k + h_0) \le \lambda^*(\boldsymbol{x}) < (b_{k+1} + h_0)$, and*

$$z_j^*(z_0, \boldsymbol{x}) = \begin{cases} 0, & \text{if } j \le k; \\ F_j^{-1}\left(\dfrac{b_j + h_0 - \lambda^*(\boldsymbol{x})}{b_j + h_j'}\middle|\boldsymbol{x}_j\right), & \text{if } j > k, \end{cases} \tag{12}$$

where $\lambda^*(\boldsymbol{x})$ satisfies $\sum_{j=1}^J F_j^{-1}\left(\frac{b_j + h_0 - \lambda^*(\boldsymbol{x})}{b_j + h'_j}\middle|\boldsymbol{x}_j\right) = z_0$.

Figure 1 illustrates Proposition 1 on how the optimal allocated quantities change according to z_0. Specifically, one can view a_k as a threshold that indicates which retailer receives a positive inventory allocation when moving z_0 from large to small. For example, when z_0 is larger than $\sum_{i=1}^J s_i^e$, the warehouse has sufficient inventory to meet every retailer j's base-stock level s_j^e; when $z_0 \in [a_1, \sum_{i=1}^J s_i^e)$, every retailer j can still receive positive inventory allocation according to (11). When z_0 continues to decrease to the region $[a_2, a_1)$, retailer 1 will not receive any inventory (i.e., $z_1^*(z_0, \boldsymbol{x}) = 0$) as it has the smallest lost-sales penalty b_1. The warehouse inventory will be allocated to the remaining retailers according to (12); that is, the marginal inventory cost of post allocation is the same between the remaining retailers. The same idea applies to the other intervals.

We now turn to the first-stage problem, which can be reformulated as

$$\min_{z_0 > 0} h_0 \cdot z_0 + \mathbb{E}_{\boldsymbol{x}}\left[\sum_{j=1}^J C_j(z_j^*(z_0, \boldsymbol{x})|\boldsymbol{x}_j)\right]. \tag{13}$$

It is straightforward to see that (13) is convex in z_0. However, unlike the problem without features in the classic model, obtaining the optimal z_0 with features suffers from the curse of dimensionality, as one needs to take expectation over $\boldsymbol{x}$ to compute the retailer's expected cost. In addition, as $z_j^*(z_0, \boldsymbol{x})$ is a function of feature $\boldsymbol{x}$, it is also time-consuming to search for the optimal solutions of $\lambda^*(\boldsymbol{x})$ and z_0. To resolve this issue, we propose a heuristic based on Taylor approximation in the next section.

4.2. Taylor Approximation

We apply the Taylor expansion to approximate the cost function $C_j(z_j|\boldsymbol{x}_j)$ for each retailer. Recall $s_j^l(\boldsymbol{x}_j)$ in (7). As before, we abbreviate it as s_j^l. Note that $s_j^l \le s_j^e$ for all $j \in [J]^+$. We construct the second-degree Taylor approximation $\tilde{C}_j(z_j|\boldsymbol{x}_j)$ of the newsvendor cost $C_j(z_j|\boldsymbol{x}_j)$ in (5) at s_j^l as follows:

$$\begin{aligned} \tilde{C}_j(z_j|\boldsymbol{x}_j) &\triangleq C_j(s_j^l|\boldsymbol{x}_j) + C_j'(s_j^l|\boldsymbol{x}_j)(z_j - s_j^l) \\ &\quad + C_j''(s_j^l|\boldsymbol{x}_j)\frac{(z_j - s_j^l)^2}{2} \\ &= C_j(s_j^l|\boldsymbol{x}_j) - h_0(z_j - s_j^l) \\ &\quad + \frac{(b_j + h_j')f_j(s_j^l|\boldsymbol{x}_j)}{2}(z_j - s_j^l)^2, \end{aligned}$$

where C_j' and C_j'' denote the first and second derivatives with respect to z_j, and the second equality holds because of $C_j'(s_j^l|\boldsymbol{x}_j) = -h_0$ and $C_j''(s_j^l|\boldsymbol{x}_j) = (b_j + h_j')f_j(s_j^l|\boldsymbol{x}_j)$.

Replacing $C_j(z_j|\boldsymbol{x}_j)$ with $\tilde{C}_j(z_j|\boldsymbol{x}_j)$, Problem (6) can be expressed as the following quadratic optimization problem:

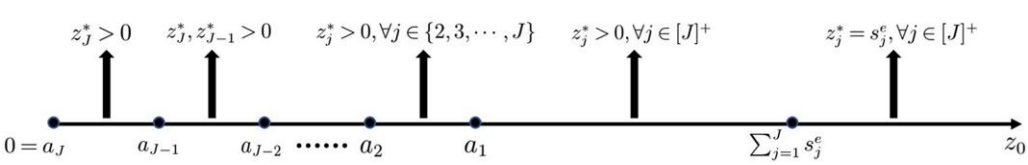
$$\begin{aligned} \tilde{z}_j(z_0, \boldsymbol{x}) \triangleq \arg\min_{z_j > 0, j \in [J]^+} \sum_{j=1}^J \Bigg[& C_j(s_j^l|\boldsymbol{x}_j) - h_0(z_j - s_j^l) \\ & + \frac{(b_j + h_j')f_j(s_j^l|\boldsymbol{x}_j)}{2}(z_j - s_j^l)^2 \Bigg] \\ \text{s.t.} \quad & \sum_{j=1}^J z_j \le z_0. \end{aligned} \tag{14}$$

For retailer $j \in [J]^+$, we further define

$$r_j \triangleq \frac{1}{(b_j + h_j')f_j(s_j^l|\boldsymbol{x}_j)}, \quad \tilde{s}_j^e \triangleq s_j^l + h_0 \cdot r_j, \quad m_j \triangleq \frac{\tilde{s}_j^e}{r_j}, \text{ and}$$
$$\underline{m} \triangleq \min_j\{m_1, m_2, \ldots, m_J\}.$$

Here, $\tilde{s}_j^e$ can be regarded as the approximated echelon base-stock level to s_j^e under Taylor approximation.

Figure 1. Optimal Allocation Policy



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Similar to the analysis in Section 4.1, we can reindex the retailers by m_j in increasing order without loss of generality and let $\underline{m} = m_1 \le m_2 \le \cdots \le m_J$. The following proposition characterizes the corresponding allocation policy under Taylor approximation.

Proposition 2 (Allocation Policy Under Taylor Approximation). *Define $\tilde{a}_k \triangleq \sum_{i=k}^{J}(\tilde{s}_i^e - m_k \cdot r_i)$. For any retailer $j \in [J]^+$, the allocation policy $\tilde{z}_j(z_0, \boldsymbol{x})$ under Taylor approximation in (14) can be expressed as follows:*

- *When $z_0 > \tilde{a}_1$,*

$$\tilde{z}_j(z_0,\boldsymbol{x}) = \begin{cases} \tilde{s}_j^e - \dfrac{r_j}{\sum_{j=1}^{J} r_j} \cdot \left(\sum_{j=1}^{J} \tilde{s}_j^e - z_0\right), & \text{if } z_0 < \sum_{j=1}^{J} \tilde{s}_j^e; \\ \tilde{s}_j^e, & \text{if } z_0 \ge \sum_{j=1}^{J} \tilde{s}_j^e. \end{cases} \tag{15}$$

- *When $\tilde{a}_{k+1} < z_0 \le \tilde{a}_k$ for $k \in [J-1]^+$,*

$$\tilde{z}_j(z_0,\boldsymbol{x}) = \begin{cases} 0, & \text{if } j \le k; \\ \tilde{s}_j^e - \dfrac{r_j}{\sum_{j=k+1}^{J} r_j} \cdot \left(\sum_{j=k+1}^{J} \tilde{s}_j^e - z_0\right), & \text{if } j > k. \end{cases} \tag{16}$$

Proposition 2 is a parallel result to Proposition 1 when we approximate the retailers' cost functions with Taylor expansion. Compared with the exact optimal solution in Proposition 1, the Taylor approximation has a closed-form solution, which significantly simplifies the computation as one does not need to search for the optimal Lagrange multiplier $\lambda^*(\boldsymbol{x})$ for the allocation decisions.

For the first-stage decision z_0: under Taylor approximation, Problem (13) can be rewritten as

$$\min_{z_0 > 0} h_0 \cdot z_0 + \mathbb{E}_{\boldsymbol{x}}\left[\sum_{j=1}^{J} C_j(\tilde{z}_j(z_0,\boldsymbol{x}) \mid \boldsymbol{x}_j)\right].$$

Although solving this problem is simpler than the exact one in (13) (as there is no need to search for $\lambda^*(\boldsymbol{x})$), the computational time is still extensive. Here we propose a simple order policy for the warehouse based on the following observation. Note that $s_j^e(\boldsymbol{x}_j) \ge s_j^l(\boldsymbol{x}_j)$. This suggests that the maximum inventory amount allocated to retailer j in the original model is larger than that in the AF model. Thus, with the larger target base-stock levels at the downstream sites, the warehouse does not need to order the sum of the maximum levels $\sum_{j=1}^{J} s_j^e(\boldsymbol{x}_j)$ because the warehouse faces uncertain features $\boldsymbol{x}$ and can optimize the allocation decision with a smaller amount of inventory after the realization of the features. This motivates us to choose the following inventory order quantity for the warehouse:

$$\tilde{z}_0 = \mathbb{E}_{\boldsymbol{x}}\left[\sum_{j=1}^{J} F_j^{-1}\left(\frac{b_j}{b_j + h_j'} \middle| \boldsymbol{x}_j\right)\right] = \mathbb{E}_{\boldsymbol{x}}\left[\sum_{j=1}^{J} s_j^l(\boldsymbol{x}_j)\right]. \tag{17}$$

Computing $\tilde{z}_0$ in (17) is rather simple as we only need to take the expectation over $\boldsymbol{x}_j$ on s_j^l. We refer to $(\tilde{z}_0, \tilde{z}_j(\tilde{z}_0, \boldsymbol{x})), j \in [J]^+$ as the *Taylor Approximation* policy. As shown in the following subsection, the TA policy is asymptotically optimal when the number of retailers J approaches infinity.

4.3. Performance Guarantee of the TA Policy

We provide theoretical bounds for the performance of the TA policy. We first provide the performance bounds for the allocation decision for a given z_0 under the TA policy. This result leads to the main finding that the TA policy is asymptotically optimal when J approaches infinity. We also provide a probabilistic performance bound for given demand features $\boldsymbol{x}$ under the TA policy. The following assumptions will be used in the rest of Sections 4 and 5 to establish our technical results.

Assumption 1 (Linear Demand Model). *For all $j \in [J]^+$, the demand of each retailer j follows*

$$D_j = \gamma_j + \sum_{p=1}^{P} \beta_{jp} \cdot x_{jp} + \epsilon_j,$$

where γ_j is a constant, $\boldsymbol{\beta}_j = (\beta_{j1}, \ldots, \beta_{jP})$ is the nonzero linear coefficient vector with $\|\boldsymbol{\beta}_j\|_\infty \le \overline{B}$, and ϵ_j is a bounded random noise with mean zero, independent of feature $\boldsymbol{x}_j$ and over retailer $j \in [J]^+$.

The linear demand assumption is widely used in the literature as it is simple and representative (see Ban and Rudin 2019). Here, we assume all retailers have feature vectors with the same dimension for notational simplicity. Our results still hold for retailers with a nonidentical number of features P_j.

This assumption makes s_j^l a linear function of features:

$$s_j^l = F_j^{-1}\left(\frac{b_j}{b_j + h_j'} \middle| \boldsymbol{x}_j\right) = \beta_{j0} + \sum_{p=1}^{P} \beta_{jp} \cdot x_{jp}, \tag{18}$$

where $\beta_{j0} = \gamma_j + F_{\epsilon_j}^{-1}(b_j/(b_j + h_j'))$ and $F_{\epsilon_j}(\cdot)$ is the cumulative distribution function for random noise ϵ_j. With this, $\tilde{z}_0$ in (17) can be easily obtained.

Assumption 2 (Demand Distribution). *Demand D_j and random noise ϵ_j for each retailer $j \in [J]^+$ are continuous and*

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bounded in $[0,\overline{D}]$ *and* $[-\overline{\epsilon},\overline{\epsilon}]$, *respectively. The conditional cumulative distribution functions* $F_j(\cdot|\boldsymbol{x}_j)$ *of demand* D_j *are absolutely continuous and strictly increasing, and the density function* $f_j(\cdot|\boldsymbol{x}_j)$ *is Lipschitz continuous and uniformly bounded on its domain for any* $\boldsymbol{x}_j$. *In particular, there exist two constants* $0 < \ell_j \le u_j < \infty$ *that* $f_j(\cdot|\boldsymbol{x}_j) \in [\ell_j, u_j]$ *for any* $\boldsymbol{x}_j$ *and* $j \in [J]^+$.

The above assumptions are widely used in the inventory literature (see Ban and Rudin 2019, Ding et al. 2021, Keskin et al. 2021). Bounded demand is a prerequisite condition for a bounded inventory cost. The strictly increasing property for $F_j(\cdot|\boldsymbol{x}_j)$ is to ensure the uniqueness of the local base-stock level s_j^l. The Lipschitz continuous property of $f_j(\cdot|\boldsymbol{x}_j)$ is used in Section 5 to prove the consistency result of our data-driven solution. The upper and lower bounds for the conditional probability density function (pdf) are used to guarantee the performance of our TA policy compared with the optimal policy, as we will show later in Proposition 3. We note that for any given $\boldsymbol{x}_j$, the conditional CDF and pdf of demand D_j are equivalent to those of random noise ϵ_j (with a shift in the x-scale) for all $j \in [J]^+$.

Assumption 3 (Demand Feature). *Features* x_{jp} *are independent and bounded random variables with two constants* $\overline{M}$ *and* $\overline{\sigma}$ *such that* $|x_{jp}| \le \overline{M}$ *and* $\mathrm{Var}(x_{jp}) \le \overline{\sigma}^2$ *for all* $j \in [J]^+$ *and* $p \in [P]^+$.

The bounded demand feature setting here is mainly deduced from Assumption 1 and Assumption 2, because our demand and random noise are all bounded random variables. Bounded distribution is a special class of light-tailed distributions, and our results can also be extended to the case with light-tailed demand/feature random variables, based on Qi et al. (2020), who study the data-driven newsvendor problem with demand/feature following light-tailed distributions.

Given the warehouse's inventory level z_0 and realized demand features $\boldsymbol{x}$, we define the difference between the cost under Taylor approximation solutions and the optimal cost as follows:

$$\tau(z_0,\boldsymbol{x}) \triangleq \sum_{j=1}^{J}[C_j(\tilde{z}_j(z_0,\boldsymbol{x})|\boldsymbol{x}_j) - C_j(z_j^*(z_0,\boldsymbol{x})|\boldsymbol{x}_j)].$$

Define $t_j \triangleq 1/((b_j+h_j')\cdot\ell_j)$ and $\overline{s}_j^e \triangleq s_j^l + h_0\cdot t_j$. Recall the definitions of r_j and $\tilde{s}_j^e$ for the TA policy, and we have $r_j \le t_j$ and $\tilde{s}_j^e \le \overline{s}_j^e$. The following proposition gives the performance bound of the allocation decision under the TA policy with given z_0.

Proposition 3 (Performance Bound on Allocation). *Under Assumption 2, the allocation decision of the TA policy in (15) has the following performance bound. For given* $\boldsymbol{x}$ *and any* $z_0 > \sum_{j=1}^{J}(\tilde{s}_j^e - \underline{m}\cdot r_j)$,[5]

$$\tau(z_0,\boldsymbol{x}) \le \begin{cases} (\alpha^3-1)\dfrac{(\sum_j s_j^l - z_0)^2}{2\sum_j t_j}, \\ \qquad\qquad \text{if } \displaystyle\sum_{j=1}^{J}(\tilde{s}_j^e - \underline{m}\cdot r_j) < z_0 < \sum_{j=1}^{J}\tilde{s}_j^e; \\ \dfrac{\alpha-1}{\alpha}h_0\left(\displaystyle\sum_j t_j\right) + \dfrac{\alpha^3-1}{2\alpha^2}(h_0)^2\left(\displaystyle\sum_j t_j\right), \\ \qquad\qquad \text{if } \displaystyle\sum_{j=1}^{J}\tilde{s}_j^e \le z_0 < \sum_{j=1}^{J}\overline{s}_j^e; \\ \dfrac{\alpha-1}{\alpha}h_0\left(\displaystyle\sum_j t_j\right) + \dfrac{\alpha-1}{2}(h_0)^2\left(\displaystyle\sum_j t_j\right), \\ \qquad\qquad \text{if } z_0 \ge \displaystyle\sum_{j=1}^{J}\overline{s}_j^e; \end{cases}$$

where $\alpha = \max_j\{u_j/\ell_j\}$ *represents the maximum ratio between the upper and lower bounds of conditional probability density functions.*

Proposition 3 suggests the following two results. First, if $\forall j$ the random noise ϵ_j follows a uniform distribution, then $\alpha = 1$, and the allocation decision under the TA policy is optimal. Second, the allocation decision performs more effectively when z_0 is closer to $\sum_{j=1}^{J} s_j^l$. This is intuitive because the Taylor expansion closely approximates the exact retailer's cost function in the neighborhood of the expansion point s_j^l.

Let π denote any feasible policy with order quantity z_0^π and allocation quantity $z_j^\pi(z_0^\pi,\boldsymbol{x})$ for any given feature $\boldsymbol{x}$ for all $j \in [J]^+$. The conditional total inventory cost and the expected inventory cost over feature $\boldsymbol{x}$ under policy π are defined as

$$\mathbf{C}_\pi(\boldsymbol{x}) \triangleq h_0\cdot z_0^\pi + \sum_{j=1}^{J} C_j(z_j^\pi(z_0^\pi,\boldsymbol{x})|\boldsymbol{x}_j), \text{ and}$$
$$\mathbf{C}_\pi \triangleq \mathbb{E}_{\boldsymbol{x}}[\mathbf{C}_\pi(\boldsymbol{x})]. \tag{19}$$

Specifically, $\pi = \mathbf{OPT}$ denotes the optimal policy, and $\pi = \mathbf{TA}$ denotes the Taylor approximation policy, with their order quantities abbreviated as z_0^* and $\tilde{z}_0$, respectively. The following theorem shows the asymptotic optimality of the TA policy.

Theorem 1 (Asymptotic Optimality of the TA Policy). *Under Assumptions 1–3, the following results hold:*

(i) $(\mathbf{C_{TA}} - \mathbf{C_{OPT}})/\mathbf{C_{OPT}} = O(J^{-1})$, $\tilde{z}_0 - \mathbb{E}_{\boldsymbol{x}}[\sum_{j=1}^{J} s_j^l] = 0$, *and* $\mathbb{E}_{\boldsymbol{x}}[|\tilde{z}_j - s_j^l|] = O(1)$.

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(ii) *For any given feature* $\boldsymbol{x}$, *there exists a constant* $c_1 > 0$ *such that for any* $0 < \delta < 1 - 2\exp(-c_1 J)$, *with probability* $1 - 2\exp(-c_1 J) - \delta$ *we have* $(\mathbf{C_{TA}}(\boldsymbol{x}) - \mathbf{C_{OPT}}(\boldsymbol{x}))^+ / \mathbf{C_{OPT}}(\boldsymbol{x}) = O(J^{-1}\ln 2/\delta + \sqrt{J^{-1}\ln 2/\delta})$. *Moreover, with probability* $1-\delta$, *we have* $|\tilde{z}_0 - \sum_{j=1}^J s_j^l| = O(\sqrt{J \ln 2/\delta})$ *and with probability* $1 - 2\exp(-c_1 J) - \delta$, $|\tilde{z}_j - s_j^l| = O(\sqrt{J^{-1} \ln 2/\delta})$.

Theorem 1 demonstrates that the TA policy is asymptotically optimal when the number of retailers J approaches infinity. The idea of proving Theorem 1 is to compare the TA policy with the policy under the AF system described in Section 3 because it is difficult to compute the exact optimal cost $\mathbf{C_{OPT}}$. One can show that optimality gap $(\mathbf{C_{TA}} - \mathbf{C_{OPT}}) \le (\mathbf{C_{TA}} - \mathbf{C_{AF}}) = O(1)$, regardless of the size of J. The result suggests that the warehouse's order quantity can be as low as $\tilde{z}_0$ to achieve the asymptotic optimality. It is possible that one can construct a heuristic policy where the warehouse orders an amount of $\tilde{z}_0 + O(g(J))$, with $g(J) > 0$ and $\lim g(J) = \infty$ to ensure that the warehouse can ship $s_j^l(\boldsymbol{x}_j)$ with high probability when J is large, making the allocation decision less important. Nevertheless, the gap between the cost of such a policy and $\mathbf{C_{OPT}}$ is at least $O(g(J))$, making the convergence rate slower than the TA policy. This demonstrates the importance of the allocation decision. We refer the reader to Online Appendix B.1 for a detailed discussion.

5. Data-Driven Inventory Policy

In this section, we extend our previous analysis to the case where the distributions are unknown and each retailer $j \in [J]^+$ has historical data $\{(d_j^i, \boldsymbol{x}_j^i)\}_{i=1}^N$. We shall illustrate how to generate a data-driven policy based on the TA policy.

Our idea is to use statistical learning algorithms to substitute the key components in the TA policy. We refer to the resulting policy as the *Data-Driven Taylor Approximation* policy. Following Proposition 2, we need to use the historical data to calculate the retailer's base-stock level $\tilde{s}_j^e$ (or, equivalently, s_j^l) and the probability density function $f_j(s_j^l|\boldsymbol{x}_j)$ for given $\boldsymbol{x}$. We adopt the linear quantile regression, kernel density estimation, and sample average to generate the estimators of $s_j^l, f_j(s_j^l|\boldsymbol{x}_j)$, and $\tilde{z}_0$ in Sections 5.1–5.3, respectively. In Section 5.4, we show the effectiveness of the DDTA policy by characterizing the asymptotic performance bound on the estimation error $|\mathbf{C_{DDTA}}(S_N) - \mathbf{C_{TA}}|$ in Theorem 2 and the generalization bound $|\mathbf{R_{DDTA}}(S_N) - \mathbf{C_{DDTA}}(S_N)|$ in Theorem 3.

5.1. Base-Stock Level Estimation

To estimate $\tilde{s}_j^e$, we need to estimate s_j^l first. Based on the linear demand model, the local base-stock level s_j^l for each retailer j is the linear function of demand features, that is, $s_j^l = \beta_{j0} + \sum_{p=1}^P \beta_{jp} x_{jp}$. We can use the training (in-sample) data set to obtain the linear coefficients through the empirical risk minimization approach for each retailer $j \in [J]^+$ as follows:

$$\hat{\boldsymbol{\beta}}_j = \arg\min_{\boldsymbol{\beta}_j} \frac{1}{N}\sum_{i=1}^N \left[h_j' \cdot \left(\beta_{j0} + \sum_{p=1}^P \beta_{jp} x_{jp}^i - d_j^i\right)^+ + b_j \cdot \left(d_j^i - \beta_{j0} - \sum_{p=1}^P \beta_{jp} x_{jp}^i\right)^+ \right]. \tag{20}$$

Problem (20) is the same as the linear quantile regression which estimates the quantile for a given $\boldsymbol{x}_j$. The solution $\hat{s}_j^l \triangleq \hat{\beta}_{j0} + \sum_{p=1}^P \hat{\beta}_{jp} x_{jp}$ is actually demand D_j's $b_j/(b_j + h_j')$-th estimated quantile conditional on $\boldsymbol{x}_j$. Several papers have shown that the estimated quantile $\hat{s}_j^l$ will converge to true quantile s_j^l when N approaches infinity, for example, Koenker (2005), Ban and Rudin (2019), and Qi et al. (2022).

5.2. Kernel Density Estimation

To estimate the probability density function $f_j(s_j^l|\boldsymbol{x}_j)$ for given $\boldsymbol{x}_j$ for $j \in [J]^+$, we adopt the commonly used kernel density estimation method in nonparametric statistics. The formula can be expressed as

$$\hat{f}_j(\hat{s}_j^l|\boldsymbol{x}_j) = \frac{1}{N} \cdot \sum_{i=1}^N \frac{1}{w_N} K\left(\frac{\hat{s}_j^l - d_j^i}{w_N}\right) = \frac{1}{N} \cdot \sum_{i=1}^N \frac{1}{w_N} K\left(\frac{\hat{\beta}_{j0} + \sum_{p=1}^P \hat{\beta}_{jp} x_{jp} - d_j^i}{w_N}\right), \tag{21}$$

where w_N is the bandwidth satisfying $\lim_{N\to\infty} w_N = 0$ and $\lim_{N\to\infty} N w_N = \infty$. $K(\cdot)$ is set to be the Gaussian kernel defined as $K(x) \triangleq \exp(-x^2/2)/\sqrt{2\pi}$. Gaussian kernel is one of the most widely used kernel functions in nonparametric statistics. One can use other kernel functions which satisfy the conditions of being symmetric and compactly supported, with Hölder-continuous probability densities and $K(0) \neq 0$, like the Epanechnikov kernel, Tricube kernel, etc., but what affects the accuracy of estimation most is the selection of bandwidth. Following theorem 2 of Jiang (2017), we determine the bandwidth as $w_N = a_0 N^{-1/3}$, where $a_0 > 0$ is the coefficient that minimizes the cross-validation loss in the data set.

5.3. Data-Driven Taylor Approximation

With the estimation on the base-stock levels $\hat{s}_j^l$ and conditional probability functions $\hat{f}_j(\hat{s}_j^l|\boldsymbol{x}_j)$ in Sections 5.1 and 5.2, we now provide the DDTA policy.

Define $\hat{r}_j \triangleq \frac{1}{(b_j + h_j')\hat{f}_j(\hat{s}_j^l|\boldsymbol{x}_j)}$, $\hat{s}_j^e \triangleq \hat{s}_j^l + h_0 \cdot \hat{r}_j$, $\hat{m}_j \triangleq \frac{\hat{s}_j^e}{\hat{r}_j}$, and $\underline{\hat{m}} = \min_j \hat{m}_j$. Following Proposition 2, we also reindex the retailer j by $\hat{m}_1 \le \hat{m}_2 \le \cdots \le \hat{m}_J$ and define

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$\hat{a}_k \triangleq \sum_{i=k}^{J}(\hat{s}_i^e - \hat{m}_k \cdot \hat{r}_i)$. The data-driven allocation decision $\hat{z}_j(z_0, \boldsymbol{x})$ for any retailer $j \in [J]^+$ can be expressed as follows:

- When $z_0 > \hat{a}_1$,

$$\hat{z}_j(z_0,\boldsymbol{x}) = \begin{cases} \hat{s}_j^e - \dfrac{\hat{r}_j}{\sum_{j=1}^{J}\hat{r}_j}\cdot\left(\sum_{j=1}^{J}\hat{s}_j^e - z_0\right), & \text{if } z_0 < \sum_{j=1}^{J}\hat{s}_j^e; \\ \hat{s}_j^e, & \text{if } z_0 \geq \sum_{j=1}^{J}\hat{s}_j^e; \end{cases} \tag{22}$$

- When $\hat{a}_{k+1} < z_0 \leq \hat{a}_k$ for $k \in [J-1]^+$,

$$\hat{z}_j(z_0,\boldsymbol{x}) = \begin{cases} 0, & \text{if } j \leq k; \\ \hat{s}_j^e - \dfrac{\hat{r}_j}{\sum_{j=k}^{J}\hat{r}_j}\cdot\left(\sum_{j=k}^{J}\hat{s}_j^e - z_0\right), & \text{if } j > k. \end{cases} \tag{23}$$

For the warehouse's replenishment decision, we use the sample average to estimate the expectation in (17), which can be written as

$$\hat{z}_0 = \frac{1}{N}\cdot\sum_{i=1}^{N}\sum_{j=1}^{J}\hat{s}_j^l(\boldsymbol{x}_j^i) = \frac{1}{N}\cdot\sum_{i=1}^{N}\sum_{j=1}^{J}\left[\hat{\beta}_{j0} + \sum_{p=1}^{P}\hat{\beta}_{jp}x_{jp}^i\right]. \tag{24}$$

5.4. Performance Analysis for the DDTA Policy

Theorem 1 in Section 4 has already shown the bound on optimality gap. In this subsection, we aim to provide the analysis of the estimation error (see Theorem 2) and generalization bound (see Theorem 3). Combining these results can yield the performance bound on how well our in-sample inventory cost under the DDTA policy can approximate the optimal expected total inventory cost. We need mild assumptions on the data set to derive these results.

Assumption 4 (Sample Process). *For independent and identically sampled historical data $S_N \triangleq \{(d_1^i, \boldsymbol{x}_1^i), (d_2^i, \boldsymbol{x}_2^i), \ldots, (d_J^i, \boldsymbol{x}_J^i)\}_{i=1}^N$, there exist positive definite matrices M_j^0 and a positive constant $\overline{M} > 0$ such that almost surely for any $j \in [J]^+$,*

(i) $\lim_{N\to\infty}\frac{1}{N}\cdot\sum_{i=1}^{N}\boldsymbol{x}_j^{i\top}\boldsymbol{x}_j^i = M_j^0$;

(ii) $\max_{i=1,\ldots,N}\|\boldsymbol{x}_j^i\|_\infty \leq \overline{M}$.

This assumption is widely used in regression fields to develop the asymptotic property (see Koenker 2005, Ban et al. 2019, Qi et al. 2022). Part (i) guarantees the existence of inverse matrix $\left(\sum_{i=1}^{N}\boldsymbol{x}_j^{i\top}\boldsymbol{x}_j^i/N\right)^{-1}$, which further ensures the existence of $\hat{\boldsymbol{\beta}}$ for N sufficiently large almost surely. Part (ii) states that all $\boldsymbol{x}_j^i$ have a finite sup norm, which is consistent with Assumption 3.

The following proposition shows that both the allocation decision $\hat{z}_j(z_0, \boldsymbol{x})$ and the warehouse's replenishment decision $\hat{z}_0$ converge in probability to the corresponding solutions under the TA policy, and it provides the convergence rates. We use $O_p(1)$ and $o_p(1)$ to denote a sequence of random variables which are bounded in probability and converging in probability to zero, respectively. Further, $W_n = O_p(R_n)$ and $W_n = o_p(R_n)$ represent $W_n = Y_n R_n$ with $Y_n = O_p(1)$ and $Y_n = o_p(1)$, respectively.

Proposition 4 (Convergence Rates of the DDTA Policy). *Under Assumptions 1, 2, and 4,*

(i) *for any features $\boldsymbol{x}$ and z_0, the allocation decision of the DDTA policy in (22)–(23) converges in probability to that of the TA policy in (15)–(16) with the convergence rate $|\hat{z}_j(z_0,\boldsymbol{x}) - \tilde{z}_j(z_0,\boldsymbol{x})| = o_p(PN^{-1/3})$;*

(ii) *the replenishment decision of the DDTA policy in (24) converges in probability to that of the TA policy in (17) with the convergence rate $|\hat{z}_0 - \tilde{z}_0| = O_p(JPN^{-1/2})$.*

Proposition 4 shows that the allocation and replenishment decisions under the DDTA policy converge to those under the TA policy with a probability of one for sufficiently large N, and characterizes the convergence rates of the predictors. In particular, $|\hat{z}_j(z_0,\boldsymbol{x}) - \tilde{z}_j(z_0,\boldsymbol{x})|$ converges in probability to zero with a rate faster than $PN^{-1/3}$. The same interpretation applies to $\hat{z}_0$ and $\tilde{z}_0$.

For policy $\pi = \mathbf{DDTA}$, according to (19), define $\mathbf{C_{DDTA}}(\boldsymbol{x}, S_N) \triangleq h_0 \cdot \hat{z}_0 + \sum_{j=1}^{J} C_j(\hat{z}_j(\hat{z}_0,\boldsymbol{x})|\boldsymbol{x}_j)$ as the expected total cost over random noises under the data-driven solution for any out-of-sample features $\boldsymbol{x}$, with the DDTA policy $\hat{z}_0$ and $\hat{z}_j(\hat{z}_0,\boldsymbol{x})$ trained on the data set S_N. Define $\mathbf{C_{DDTA}}(S_N) \triangleq \mathbb{E}_{\boldsymbol{x}}[\mathbf{C_{DDTA}}(\boldsymbol{x}, S_N)]$, which represents the expected out-of-sample inventory cost under the DDTA policy trained on S_N, that is, the generalization cost. By Proposition 4, we have following convergence results for the estimation error $|\mathbf{C_{DDTA}}(S_N) - \mathbf{C_{TA}}|$:

Theorem 2 (Convergence Rate for Estimation Error). *Under Assumptions 1, 2, and 4,*

(i) *for any given features $\boldsymbol{x}$ and data set S_N, $\mathbf{C_{DDTA}}(\boldsymbol{x}, S_N)$ converges in probability to $\mathbf{C_{TA}}(\boldsymbol{x})$ with the convergence rate $|\mathbf{C_{DDTA}}(\boldsymbol{x}, S_N) - \mathbf{C_{TA}}(\boldsymbol{x})| = o_p(J^2PN^{-1/3})$;*

(ii) *the expected total cost $\mathbf{C_{DDTA}}(S_N)$ converges in probability to $\mathbf{C_{TA}}$ with the convergence rate $|\mathbf{C_{DDTA}}(S_N) - \mathbf{C_{TA}}| = o_p(J^2PN^{-1/3})$.*

Theorem 2 states that the expected inventory cost under the DDTA policy converges to that of the TA policy in probability when the sample size approaches infinity with a rate faster than $J^2PN^{-1/3}$. To ensure the convergence results, we require N to be a function of J and P. For example, if J becomes 10 times larger, N needs to be 10^6 times larger to ensure the same level of convergence.

Because $\mathbf{C_{DDTA}}(S_N)$ also relies on the unknown distribution information, it is often approximated by the

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in-sample average loss defined as

$$\mathbf{R_{DDTA}}(S_N) \triangleq \frac{1}{N}\sum_{i=1}^{N}\Bigg[h_0\cdot\hat{z}_0+\sum_{j=1}^{J}[-h_0\cdot d_j^i+h_j\cdot(\hat{z}_j(\hat{z}_0,\boldsymbol{x}^i) - d_j^i)+(b_j+h_j')\cdot(d_j^i-\hat{z}_j(\hat{z}_0,\boldsymbol{x}^i))^+]\Bigg], \tag{25}$$

where the replenishment decision $\hat{z}_0$ is uniform for all instances in S_N and $\hat{z}_j(\hat{z}_0,\boldsymbol{x}^i)$ stands for the allocation decision for retailer j given $\hat{z}_0$ and the i-th instance of feature vector $\boldsymbol{x}$ in S_N. $\mathbf{R_{DDTA}}(S_N)$ is also known as empirical cost and can be calculated from the data set S_N. The following theorem characterizes the difference between the generalization cost and the empirical cost, that is, $|\mathbf{R_{DDTA}}(S_N)-\mathbf{C_{DDTA}}(S_N)|$, referred to as the generalization bound.

Theorem 3 (Generalization Bound for the DDTA Policy). *Under Assumptions 1, 2, and 4, for any $N\geq 1$ and $0<\delta<1$, there exists a constant $c_0>0$ such that with probability at least $1-\delta$,*

$$|\mathbf{R_{DDTA}}(S_N)-\mathbf{C_{DDTA}}(S_N)| \leq c_0\left(JPN^{-1/3}\ln N\ln(1/\delta)+J\sqrt{N^{-1}\ln(1/\delta)}\right). \tag{26}$$

Theorem 3 provides a finite sample guarantee for our empirical cost to approximate the generalization (out-of-sample) cost, which converges to zero when N is sufficiently large. To derive the result, we prove the "uniform stability" property of our DDTA policy; see Online Appendix A for illustration.

Combining results in Theorems 1–3, we conclude that under large sample size N, the gap of $|\mathbf{R_{DDTA}}(S_N)-\mathbf{C_{OPT}}|$ can be bounded by a constant with high probability. It is thus reasonable and effective to estimate $\mathbf{C_{OPT}}$ by our in-sample costs under the DDTA policy.

6. Extensions

We discuss several extensions of our model. Specifically, we generalize the model with correlated demand features, general lead times, and censored demand data. We illustrate the corresponding TA and DDTA policies for each extension and show the results parallel to Theorems 1–3. We leave detailed analysis, discussions, and proofs to Online Appendix B.

6.1. Time-Correlated Demand Features

In practice, demand features may be correlated between time periods, for example, weather. Here, we consider feature processes following Assumption 5 below.

Assumption 5 (Correlated Feature Between Periods). *The feature sequence $\{x_{jp}^t\}_{t=1}^T$ is (strictly) stationary, and each element x_{jp}^t is a bounded random variable with two constants $\overline{M}$ and $\overline{\sigma}$ such that $|x_{jp}^t|\leq\overline{M}$ and $\mathrm{Var}(x_{jp}^t)\leq\overline{\sigma}^2$ for all $j\in[J]^+$ and $p\in[P]^+$.*

The (strictly) stationary property implies that x_{jp}^t has the same distribution for all $t\in[T]^+$; see Chapter 6.1 of Durrett (2019). Under Assumption 5, the feature x_{jp}^{t+1} at period $t+1$ may depend on the historical realization of features $\boldsymbol{x}_{jp}^{[1,t]}\triangleq(x_{jp}^1,x_{jp}^2,\ldots,x_{jp}^t)$ from period 1 to period t for any $t\in[T]^+$. This dependence will make the order and allocation decisions more involved than those of the i.i.d. feature case. In particular, the features $\boldsymbol{x}^{t+1}$ that determine the retailers' demands and the allocation decisions can be partially inferred by the observed features $\boldsymbol{x}^{[1,t]}$. As a result, sensible order and allocation decisions should include the correlation effect.

We define a policy π that incorporates the correlation effect as follows: the inventory order decision $z_0^\pi(\boldsymbol{x}^{[1,t]})$ and the allocation decision $z_j^\pi(z_0^\pi(\boldsymbol{x}^{[1,t]}),\boldsymbol{x}^{t+1})$. The expected total inventory cost in period $t+1$ under policy π conditional on $\boldsymbol{x}^{[1,t]}$ is

$$\mathbf{C}_\pi^{t+1}(\boldsymbol{x}^{[1,t]})\triangleq\mathbb{E}_{\boldsymbol{x}^{t+1}|\boldsymbol{x}^{[1,t]}}\Bigg[h_0\cdot z_0^\pi(\boldsymbol{x}^{[1,t]}) +\sum_{j=1}^{J}C_j(z_j^\pi(z_0^\pi(\boldsymbol{x}^{[1,t]}),\boldsymbol{x}^{t+1})|\boldsymbol{x}_j^{t+1})\Bigg]. \tag{27}$$

To specify the relationship between $\boldsymbol{x}^{[1,t]}$ and $\boldsymbol{x}^{t+1}$, we assume the following feature transition function for the subsequent analysis.

Assumption 6 (Feature Transition). *For any $j\in[J]^+$, $p\in[P]^+$, $t\in[T]^+$, there exist bounded functions $k_{jp}^t(\cdot):\mathbb{R}^t\to\mathbb{R}$ such that feature transition follows*

$$x_{jp}^{t+1}=k_{jp}^t(\boldsymbol{x}_{jp}^{[1,t]})+\delta_{jp}^{t+1}, \tag{28}$$

where δ_{jp}^{t+1} denotes the bounded random noise with mean zero (arising from feature transition) independently distributed over $j\in[J]^+$ and $p\in[P]^+$.

The transition function in (28) includes time series functions. For example, an AR(1) feature transition process can be expressed as $x_{jp}^{t+1}=\eta_{jp}x_{jp}^t+\delta_{jp}^{t+1}$, with $k_{jp}^t(\boldsymbol{x}_{jp}^{[1,t]})\triangleq\eta_{jp}x_{jp}^t$ and $|\eta_{jp}|<1$ being the autoregressive coefficient. Define the TA policy with time-correlated features as $\pi=\mathbf{TA^{TC}}$. For given historical feature $\boldsymbol{x}^{[1,t]}$, the replenishment decision at period t is $\tilde{z}_0^{TC}(\boldsymbol{x}^{[1,t]})\triangleq\sum_{j=1}^J\beta_{j0}+\sum_{p=1}^P\beta_{jp}\cdot k_{jp}^t(\boldsymbol{x}_{jp}^{[1,t]})$, and the allocation decision at period $t+1$ follows (15)–(16) with regard to the observed feature $\boldsymbol{x}^{t+1}$. We slightly abuse the notation to denote $\pi=\mathbf{OPT}$ the optimal policy under correlation.

Corollary 1 (Performance of $\mathrm{TA}^{\mathrm{TC}}$ Policy). *Under Assumptions 1, 2, 5, and 6 and given feature vectors $\boldsymbol{x}^{[1,t]}$, $(\mathbf{C}_{\mathbf{TA^{TC}}}^{t+1}(\boldsymbol{x}^{[1,t]})-\mathbf{C}_{\mathbf{OPT}}^{t+1}(\boldsymbol{x}^{[1,t]}))/\mathbf{C}_{\mathbf{OPT}}^{t+1}(\boldsymbol{x}^{[1,t]})=O(J^{-1})$ holds for any period $t\in[T]^+$.*

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Based on the TA^{TC} policy, we define the DDTA policy under time-correlated features (DDTA^{TC}) as follows. For the warehouse, the order quantity is $\hat{z}_0^{TC}(\boldsymbol{x}^{[1,t]}) \triangleq \sum_{j=1}^{J} \hat{\beta}_{j0} + \sum_{p=1}^{P} \hat{\beta}_{jp} \cdot \hat{k}_{jp}(x_{jp}^{[1,t]})$ and the allocation decision is the same as that of the DDTA policy (as the lead time for the retailer is zero). Here, we estimate $\boldsymbol{\beta}$ and the probability density the same way as we do in the DDTA policy, that is, viewing these feature data as generated from i.i.d. processes and applying the same machine learning approaches as those in Section 5 to generate these required components. We can show that the consistency result and generalization bound for the DDTA^{TC} policy still hold under further assumptions on $\hat{k}_{jp}(\cdot)$ (i.e., the point-wise error rate and uniform stability property) and the sampling process. The analyses follow similar steps as those in Theorems 2 and 3; see Online Appendix B.2 for details.

Although one can implement the DDTA^{TC} policy for the time-correlated feature data, it is more involved to implement and less practical. On the other hand, our TA and DDTA policies introduced in Sections 4.2 and 5.3 work well under time-correlated features. We show that the TA policy that ignores the time correlation is asymptotically optimal in J. (Although the convergence rates for the TA and TA^{TC} policy are equal to $1/J$, the constant ratio for the latter is smaller.) In addition, we can show that the DDTA policy is also consistent when N is sufficiently large under similar feature correlation conditions as in Qi et al. (2020). We refer readers to Online Appendix B.2 for a detailed discussion and proofs. In Section 8, we apply the DDTA policy for Hema's case in which demand features are time correlated.

6.2. Lead Time

The base model assumes that the warehouse's lead time is one period and the retailers' lead time is zero. In this subsection, we consider a more general setting with the warehouse's lead time $L_0 \geq 1$ and retailers' lead time $L_j \equiv L \geq 0$ for all $j \in [J]^+$. We assume that the product perishes in $L_0 + L$ periods with this setting. The inventory ordered at the beginning of period t would take L_0 periods to arrive at the warehouse, and the feature $\boldsymbol{x}^{t+L_0}$ is then observed. The allocation decision is made according to $\boldsymbol{x}^{t+L_0}$. The allocated inventory arrives at the retailer at the beginning of period $t + L_0 + L$, and extra inventory is disposed of at the end of period $t + L_0 + L$. Because of $L_0 + L$ periods of shelf life, the product which arrives at the warehouse at period $t + L_0$ either can be allocated to the retailer or perishes in the pipeline to the retailer.

When features are independent across time periods, one can show that for $L = 0$, the TA and DDTA policies can still be applied directly (i.e., with the same order and allocation decisions), and results in Theorems 1–3 still hold. For $L > 0$, the model degenerates to the no-feature model described in Section 3, with the order and allocation decisions being the minimizers of (8), as the realized features do not affect the allocation decision because of feature independence.

Note that the feature information can improve the order and allocation decisions for the general lead-time model when the features are correlated between times. This is essentially an extension of Section 6.1 with general lead times. Similar to the results in Section 6.1, we can show parallel results to Theorems 1–3. Please refer to Online Appendix B.5 for details.

6.3. Censored Demand Data

For lost-sales inventory systems, the firm may not have access to true demand data $\{(d_1^i, \ldots, d_J^i)\}_{i=1}^N$. Instead, it has records on its on-hand inventory quantity $\{(y_1^i, \ldots, y_J^i)\}_{i=1}^N$ at the beginning of each selling period (i.e., each day) and the sales data $\{(u_1^i, \ldots, u_J^i)\}_{i=1}^N$ at the end of each selling period after demand realizes, with $u_j^i = \min\{d_j^i, y_j^i\}$ for all $j \in [J]^+$ and $i \in [N]^+$. Here, we consider the offline data-driven problem by estimating the true demand from sales data. This is different from the online learning problem where the demand can be learned from the policy design over time, for example, Huh et al. (2011), Chen and Chao (2020), and Yuan et al. (2021).

In the case of demand censoring, our TA policy still works when the true demand distributions are known; that is, Theorem 1 still holds. We can follow the analysis in Section 5 to design the data-driven policy by estimating corresponding parameters. However, a direct consequence is that now the quantile estimator for the base-stock levels in Section 5.1 and the probability density estimator in Section 5.2 based on the censored data will be biased. We thus refer to statistical techniques for "recovering" the demand from sales data and deriving reliable estimators.

Consider the base-stock level estimation in (20): demand d_j^i is no longer available; instead, we have on-hand inventory y_j^i (which is not a random variable) and sales u_j^i. Following the censored linear quantile regression in Powell (1986) and Koenker (2008), we estimate the base-stock level for each retailer $j \in [J]^+$ by minimizing the following empirical in-sample inventory cost:

$$\hat{\boldsymbol{\beta}}_j^C = \arg\min_{\boldsymbol{\beta}_j} \frac{1}{N} \sum_{i=1}^{N} \left[h_j' \left(\min\left\{ y_j^i, \beta_{j0} + \sum_{p=1}^{P} \beta_{jp} x_{jp}^i \right\} - u_j^i \right)^+ + b_j \left(u_j^i - \min\left\{ y_j^i, \beta_{j0} + \sum_{p=1}^{P} \beta_{jp} x_{jp}^i \right\} \right)^+ \right], \tag{29}$$

where $\hat{\boldsymbol{\beta}}_j^C \triangleq (\hat{\beta}_{j0}^C, \hat{\beta}_{j1}^C, \ldots, \hat{\beta}_{jP}^C)$. For any given feature $\boldsymbol{x}_j$, the corresponding local base-stock level $\hat{s}_j^{lC} \triangleq \hat{\beta}_{j0}^C + \sum_{p=1}^{P} \hat{\beta}_{jp}^C x_{jp}$.

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We also need to revise the probability density estimator in (21) under demand censoring, as d_j^i is unobservable and we have only the censored sales data u_j^i. Denote $\hat{f}_j^C(\hat{s}_j^{lC}|\boldsymbol{x}_j)$ as the estimated probability density for given $\boldsymbol{x}_j$ with censored data. Following the censored kernel density estimator in Karunamuni and Yang (1991) and Khardani and Semmar (2014),

$$\hat{f}_j^C(\hat{s}_j^{lC}|\boldsymbol{x}_j) = \sum_{i=1}^{N} \frac{\delta_j^i}{N w_N \cdot \hat{G}_j(u_j^i)} \cdot K\left(\frac{\hat{\beta}_{j0}^C + \sum_{p=1}^{P} \hat{\beta}_{jp}^C x_{jp} - u_j^i}{w_N}\right), \quad (30)$$

where $K(\cdot)$, w_N stand for the Gaussian kernel function and bandwidth, respectively, as in Section 5.2, and we set $w_N = O(N^{-1/3})$. $\delta_j^i \triangleq 1(u_j^i < y_j^i)$ denotes the censoring indicator, and $1 - \hat{G}_j(t)$ is the Kaplan-Meier type estimator of cumulative demand distribution function defined as

$$\hat{G}_j(t) = \begin{cases} \prod_{i=1}^{N} \left(1 - \frac{1-\delta_j^{(i)}}{N-i+1}\right)^{\mathbf{1}(u_j^{(i)} \le t)} & \text{if } t \le u_j^{(N)}, \\ 0 & \text{otherwise,} \end{cases} \quad (31)$$

with $u_j^{(i)}$ the ordered statistics of sales data such that $u_j^{(1)} \le u_j^{(2)} \cdots \le u_j^{(N)}$. Note that we assume $\delta_j^{(N)} = 1$ for all $j \in [J]^+$ (i.e., the largest observation is uncensored) in order to avoid that the denominator in (30) equals zero. Equation (30) degenerates to the uncensored case in (21) when $\delta_j^i = 1$ for all $i \in [N]^+$.

Now we define the DDTA policy under demand censoring (DDTA^C) as follows. The allocation decision $\hat{z}_j^C(z_0, \boldsymbol{x})$ and replenishment decision $\hat{z}_0^C$ follow (22)–(24), with $\hat{s}_j^l$ and $\hat{f}_j(\cdot|\boldsymbol{x}_j)$ revised by $\hat{s}_j^{lC}$ and $\hat{f}_j^C(\cdot|\boldsymbol{x}_j)$. With these new estimators, we can derive the convergence property of the DDTA policy as in Theorem 2 under the following assumption according to Powell (1986), which is parallel to Assumption 4.

Assumption 7 (Sample from Censored Demand). *For independent and identically sampled historical data* $S_N \triangleq \{(u_1^i, \delta_1^i, \boldsymbol{x}_1^i), (u_2^i, \delta_2^i, \boldsymbol{x}_2^i), \ldots, (u_J^i, \delta_J^i, \boldsymbol{x}_J^i)\}_{i=1}^N$ *and historical inventory data* $\{y_1^i, \ldots, y_J^i\}_{i=1}^N$, *there exist positive definite matrices* M_j^0 *and a positive constant* $\overline{M} > 0$ *such that almost surely for any* $j \in [J]^+$,

(i) $\lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(\beta_{j0} + \sum_{p=1}^{P} \beta_{jp} x_{jp}^i < y_j^i) \cdot \boldsymbol{x}_j^{i\top} \boldsymbol{x}_j^i = M_j^0$, *with* $\boldsymbol{\beta}_j$ *standing for the true linear coefficients of demand's* $\frac{b_j}{b_j + h_j'}$*-th quantile.*

(ii) $\max_{i=1,\ldots,N} \|\boldsymbol{x}_j^i\|_\infty \le \overline{M}$.

For the total inventory cost under policy $\pi = \mathbf{DDTA}^C$ defined following (19), we have the following:

Proposition 5 (Rate of Convergence for Estimation Error Under Censoring). *Under Assumptions 1, 2, and 7 and that the density function* $f_j(\cdot|\boldsymbol{x}_j)$ *has a bounded second-order derivative for any* $\boldsymbol{x}_j$ *and* $j \in [J]^+$. *Then* $|\mathbf{C}_{\mathbf{DDTA}^C}(S_N) - \mathbf{C}_{\mathbf{TA}}| = O_p(J^2 P N^{-1/3})$.

The convergence rate in Proposition 5 is almost the same as that in Theorem 2. That is, the censored observation exhibits no effect on the estimator when the sample size N is sufficiently large. This is because the large data set enables us to recover true quantile coefficients $\boldsymbol{\beta}_j$ and probability density $f_j(s_j^l|\boldsymbol{x}_j)$ with the uncensored observations. However, it is still an open question how to figure out the probability bound for generalization error as in Theorem 3 with censored data.

6.4. Retailer-Correlated Demand Features

The problem becomes more involved when demand features are correlated between retailers. Unlike the time-correlated case, it is challenging to incorporate the feature correlation into the inventory order and allocation decisions. Thus, we only consider implementing the TA and DDTA policies under the retailer-correlated features. We show that the TA and DDTA policies possess similar properties as those in Theorems 1–3 when the features between retailers follow a mixing process; that is, the correlation between retailer j and $j+k$ tends to zero when $k \to \infty$; a numerical study also suggests the effectiveness of the DDTA policy in this case; see Online Appendix B.3 for detailed analysis and results. Nevertheless, we note that assuming a mixing process for retailer-correlated features is less practical. It remains an open question to design an effective data-driven inventory policy utilizing the feature correlation.

7. Numerical Study

In this section, we present a set of numerical experiments with synthesis data to quantify the value of feature information, verify the effectiveness of the TA and DDTA policies, and compare our heuristic with the other data-driven policies in the literature.

We assume that the demand of retailer j is a linear function of features

$$D_j = \gamma_j + \sum_{p=1}^{P} \beta_{jp} \cdot x_{jp} + \epsilon_j. \quad (32)$$

To capture the heterogeneity of retailers, the linear coefficients β_{jp} are sampled independently and uniformly from $(0, 1)$. The constant term is $\gamma_j = \sum_{p=1}^{P} (49 + p) \cdot \beta_{jp}$, where $P = 3$. Random features x_{jp} follow normal distributions with mean $\mu_{x_{jp}} = 0$ and variance $\sigma_{x_{jp}}^2$ and are truncated at $[\mu_{x_{jp}} - 3\sigma_{x_{jp}}, \mu_{x_{jp}} + 3\sigma_{x_{jp}}]$. Demand shocks ϵ_j also follow normal distributions with mean $\mu_{\epsilon_j} = 0$ and variance $\sigma_{\epsilon_j}^2$ with similar truncation. We fix the warehouse's local holding cost $h_0 = 1$ and assume that $1 < h_j' < b_j$ for all $j \in [J]^+$. The other system parameters will

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Table 1. The Effectiveness of the TA Policy Under Different Numbers of Retailers

Number of retailers J	2	4	8	16	32	64	128	256	512
Percentage loss ϕ_{AF}^{TA}	87.1%	40.0%	21.7%	12.4%	6.4%	2.8%	1.3%	0.7%	0.4%
Percentage loss ϕ_{OPT}^{TA}	23.9%	9.0%	6.6%	2.1%	1.4%	—	—	—	—

be specified in the subsequent experiments and can be found in Online Appendix B.6.

7.1. Value of Feature Information

We assess the value of feature information by comparing the optimal costs with and without feature information. For the no-feature model, its corresponding optimal policy is discussed in Section 3. Define $\phi_{TA}^{NF} \triangleq (\mathbf{C_{NF}} - \mathbf{C_{TA}})/\mathbf{C_{TA}} \times 100\%$ as the percentage of increased cost for not considering feature information. We find the value of feature information can be very significant. In our test, the largest ϕ_{TA}^{NF} is 228.52% and the smallest is 3.09%. In general, the value of feature information may increase in the number of retailers and in the variance of the feature and may decrease in the variance of demand shock and in the unit lost-sales cost (i.e., the critical ratio) under our examples; see Online Appendix B.6.

7.2. Asymptotic Optimality of the TA Policy

Recall in Section 4.3, we show that the TA policy is asymptotically optimal when J approaches infinity with known distributions. Here, we examine the performance of the TA policy for small values of $J = 2^i, i = 1, 2, \ldots, 9$. However, when the number of retailers is relatively large ($J \geq 32$), it is difficult to compute the optimal cost $\mathbf{C_{OPT}}$. In this situation, we compare the cost of the TA policy with that of the AF system. We define $\phi_{OPT}^{TA} \triangleq (\mathbf{C_{TA}} - \mathbf{C_{OPT}})/\mathbf{C_{OPT}} \times 100\%$ and $\phi_{AF}^{TA} \triangleq (\mathbf{C_{TA}} - \mathbf{C_{AF}})/\mathbf{C_{AF}} \times 100\%$.

From Table 1, we observe that $\mathbf{C_{TA}}$ is close to $\mathbf{C_{OPT}}$ when the number of retailers is sufficiently large, suggesting that our TA policy can be very effective. In addition, ϕ_{AF}^{TA} decreases approximately with the rate of $1/J$, which verifies Theorem 1(i).

7.3. Convergence of the DDTA Policy

We now examine the consistency of the DDTA policy when the historical demand and feature data are available. We will test the following set of primitive values: the size of training data set $N \in \{50, 100, 200, 400, 800\}$ and the number of retailers $J \in \{4, 8, 16, 32, 64\}$. Table 2 shows the empirical in-sample cost of the DDTA policy (i.e., $\mathbf{R_{DDTA}}(S_N)$, abbreviated as $\mathbf{R}$) and generalization out-of-sample cost (i.e., $\mathbf{C_{DDTA}}(S_N)$, abbreviated as $\mathbf{C}$) w.r.t. the size of the training set N and the number of retailers J. As can be seen, $\mathbf{R_{DDTA}}(S_N)$ would gradually approach $\mathbf{C_{TA}}$ as N grows.

7.4. Comparison with Other Data-Driven Policies

We compare the DDTA policy with the other three data-driven policies in the literature: the SAA, the LL method, and the PTSO method. We refer the reader to Online Appendix B.6 for a detailed description of these methods. Our goal is to examine the out-of-sample costs for these data-driven methods as well as their computational times.

Table 3 shows the generalization cost and the computational time in parentheses for each data-driven method in which the size of training data $N = 400$. Note that we do not report the training time as it can be done offline. It can be seen that the performance of the DDTA policy is very close to that of the PTSO policy and significantly better than the other two methods. In particular, SAA does not take feature information into account and thus performs the worst. The LL policy is proved to be consistent with the optimal solution when N is sufficiently large. However, its convergence rate is of order $N^{-2/(JP+2)}$, which grows exponentially with J and P. Thus, it can be observed that its performance deteriorates rapidly for fixed N when J or P becomes larger. The PTSO policy utilizes the linear demand information by first fitting a linear demand model and then conducting SAA optimization based on the estimated residuals. It can be regarded as an improved version of LL policy when the demand function is already known, which is also consistent with a rate of

Table 2. Empirical (in-Sample) Costs and Generalization (out-of-Sample) Costs Under the DDTA Policy

	$N = 50$		$N = 100$		$N = 200$		$N = 400$		$N = 800$		
	$\mathbf{R}$	$\mathbf{C}$	$\mathbf{R}$	$\mathbf{C}$	$\mathbf{R}$	$\mathbf{C}$	$\mathbf{R}$	$\mathbf{C}$	$\mathbf{R}$	$\mathbf{C}$	$\mathbf{C_{TA}}$
$J = 4$	37.44	40.46	37.14	38.47	38.63	37.31	37.62	37.01	37.08	36.51	35.80
$J = 8$	62.65	77.08	65.12	72.70	65.67	69.94	67.64	68.90	67.46	68.26	66.79
$J = 16$	113.74	150.70	120.10	141.81	121.95	135.76	123.93	133.40	124.51	131.66	128.20
$J = 32$	218.90	279.53	228.15	260.28	232.19	251.47	235.31	246.73	236.07	243.53	238.86
$J = 64$	423.82	532.54	444.34	496.61	451.92	479.72	457.07	471.07	460.27	466.84	460.09

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Table 3. Generalization Costs for Different Data-Driven Methods

	SAA	LL	PTSO	DDTA	TA	AF
$J = 4$	67.31 (–)	47.21 (0.53)	36.78 (0.48)	37.01 (0.02)	35.80	28.84
$J = 8$	147.52 (–)	114.35 (2.42)	68.32 (2.12)	68.90 (0.02)	66.79	57.06
$J = 16$	321.68 (–)	246.14 (9.44)	132.87 (7.85)	133.40 (0.04)	128.20	114.53
$J = 32$	647.67 (–)	571.20 (39.53)	246.07 (28.37)	246.73 (0.09)	238.86	225.52
$J = 64$	1,252.98 (–)	1,184.77 (198.68)	471.14 (119.79)	471.07 (0.21)	460.09	449.14

Note. Values in parentheses denote the computational time (seconds) for each instance.

convergence of order $N^{-1/2}$. The computational times for the LL and PTSO policies are close, and both are significantly larger than that of the DDTA policy, as both LL and PTSO need to solve an optimization problem with the size of N, whereas DDTA only needs to implement the closed-form decision.[6] We refer the reader to Online Appendix B.6 for a detailed comparison between DDTA and PTSO.

In the next section, we will focus on the performance of data-driven policies in Hema's case study. Hema has more than 400 "daily fresh" products, and fast computation is an important criterion to achieve same-day delivery after observing the demand features. Even for a problem of moderate size ($N = 400$, $J = 16$), it requires around an hour to generate the solutions under LL or PTSO policies, which makes these policies hard to implement in practice. In contrast, the DDTA policy is more practical with still relatively good performance, which makes it more attractive to Hema.

8. Case Study: Fresh Hema

We first describe the data set and then select features from the feature set offered by the company. The data we present here are shown on relative scales to keep the original data set confidential.

8.1. Data Description

The internal company data we collected consist of 6 months of daily inventory, sales, and feature records from October 2020 to April 2021 across the distribution system in a major city in China with 16 retail stores. We choose Chinese cabbage, labeled by Fresh Hema as "Daily Fresh," as the considered inventory item to manage. Chinese cabbage has one day of shelf life and will be disposed of at the end of each day if not sold. Hema's staff also keeps track of the in-stock period daily. With the sales amount in the in-stock period, Hema can calculate the hourly demand rate. Then, the lost sales can be estimated as the hourly demand rate times the out-of-stock period. By this procedure, Hema recovers the demand data from the censored sales data. See Figure 2 for an example of sales and demand data for Chinese cabbage at a retail store. We shall use the recovered demand data as input for the case study here.

8.2. Feature Selection

As Hema has listed more than 50 demand features in its original data set, we have to choose some of the most relevant ones in order to simplify our model. This is also recognized as the "data sparsity" issue in data-driven operation management problems; see Feng and Shanthikumar (2022) for detailed discussions. Here, we simply apply the stepwise regression used widely in statistics to select significant covariates by the Akaike information criterion (AIC). We select four features, that is, historical sales before 7 days, number of Hema members, item page views, and number of items in the shopping cart, and $P = 4$. We further conduct a hypothesis test on the data correlation, and it shows that each feature can be modeled by an ARIMA(p, d, q) process

Figure 2. Sales and Demand Data for Chinese Cabbage in a Retail Store

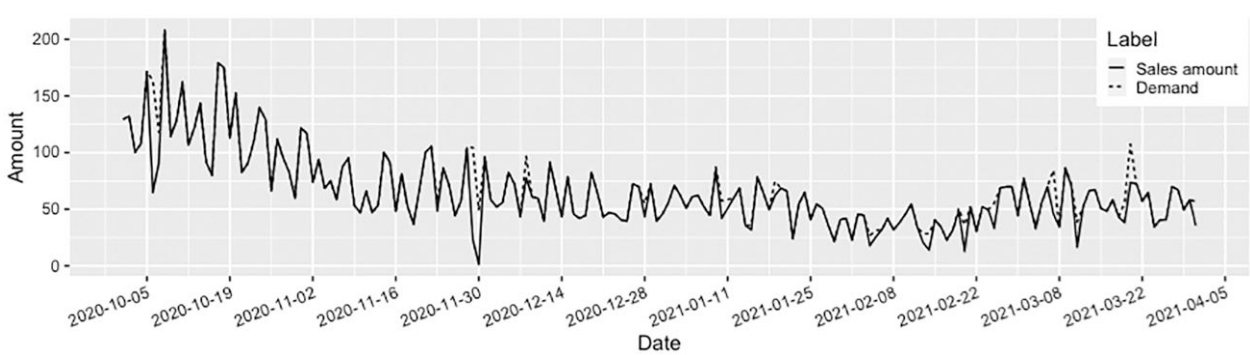

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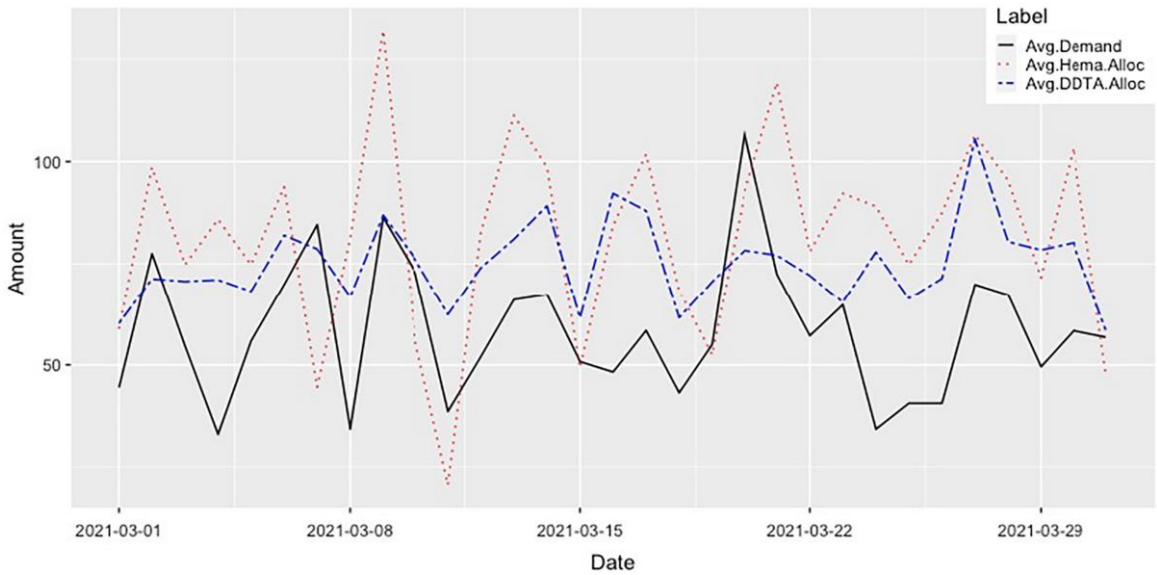
Figure 3. (Color online) Average Amount of Demand and Allocation Quantity in Each Retailer



with i.i.d. random noise. A d-order differencing method is applied to the feature sequence with $d > 0$ so that the feature data follow a stationary ARMA(p,q) process, which satisfies both Assumptions 5 and 6 in Section 6.1 and the mixing process in Assumption B8 in Online Appendix B.2; see Online Appendix B.7 for details.

8.3. Performance Comparison

As stated in Section 6.1, although we can implement the DDTA^{TC} policy for Hema's data set, the policy is more involved and difficult to use in practice. We, therefore, focus on the DDTA policy. Another reason for considering the DDTA policy is that we can fairly compare its performance with the other data-driven policies that do not utilize the feature correlation information: the SAA policy, the LL policy by Bertsimas and Kallus (2020), the PTSO policy inspired by Kannan et al. (2022), and Hema's replenishment policy (see Section 1, the second paragraph, for the description). In other words, we are implementing these policies as if the feature data were generated from i.i.d. processes, although they are time correlated.

Hema has an accounting department to estimate the cost parameters. For the considered item in this study, Hema estimates that a single unit of waste would cost $h'_j = 3.5$ and the lost-sales penalty $b_j = 30$ is nearly 10 times larger for all $j \in [J]^+$. The waste cost at the warehouse is $h_0 = 2.5$ because there are no transshipment and disposal costs. All monetary values are in Renminbi (RMB), the currency of China.

We use Hema's data from October 2020 to February 2021 to train the above four policies and use the March 2021 data as a testing set. For the LL policy, we use kNN weights and set $k = 13$, which is determined by the 10-fold cross-validation technique. We find that SAA performs poorly as the resulting cost is 89.0% higher than Hema's current policy. This demonstrates the importance of using feature information in the replenishment decision. On the other hand, LL, PTSO, and DDTA outperform the current Hema policy. Specifically, compared with Hema's policy, LL can lower the cost by 8.2%, PTSO can lower the cost by 12.0%, and DDTA can lower the cost by 11.0%. Both LL and PTSO require more than 130 seconds of total computation time, compared with DDTA, which requires only less than 0.5 seconds for the 31 instances in the test data set.

Figure 3 shows the average demand among all 16 retailers and corresponding allocation quantities under the DDTA policy and Hema's policy. (Most retailers also follow the same pattern as in the figure.) It is clear that the allocation quantities under the DDTA policy follow the demand pattern better than Hema's current policy. This is because (i) the allocation decision of the DDTA policy can reduce the order variability of the warehouse, and (ii) the DDTA policy uses the current feature information for inventory allocation whereas Hema's policy uses the day-ahead feature information for inventory replenishment, resulting in a higher variance in demand shock.

9. Concluding Remarks

This paper proposes a data-driven inventory replenishment and allocation policy for a one-warehouse, multi-retailer distribution system in which the demand for the retailer depends on features. The policy is derived from two steps. The first step is to assume distribution information is known and use Taylor expansion to approximate the retailer's cost function. We derive a

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closed-form solution, referred to as the TA policy, by solving the approximated problem. We show the TA policy is asymptotically optimal in the number of retailers. The second step is to use machine learning algorithms to generate components that require demand distributions in the TA policy based on the historical demand and feature data. We show that the resulting DDTA policy is consistent with the TA policy. Our result suggests that when the size of the sample data and the number of retailers are sufficiently large, the DDTA policy is effective. We also use a real data set obtained from Fresh Hema to compare the DDTA policy with the current one. We show that the DDTA policy can reduce the cost by 11.0% for the considered item. Finally, we extend our basic model to the settings of correlated demand features, general lead times, and censored demand data. Our proposed policy is still proven to be efficient.

There are open questions that are important and interesting for future study. First, our model considers a perishable, lost-sales distribution system with one period of lead time at the warehouse. One important research direction is to extend the current analysis to a nonperishable distribution system with backlogging. In fact, to our knowledge, the results for the backlogged system with general lead time and demand features are very limited even for single-stage models. It would be valuable to extend the analysis to provide simple and effective heuristic policies. Second, we currently study an offline problem and it would be interesting to explore how our data-driven Taylor-based heuristic could be applied to the online settings and its corresponding performance.

Endnotes

[1] Nonrandom features such as the day of the week, price, etc., can also be incorporated into our model. See Online Appendix B.4 for a brief discussion.

[2] In the multiechelon inventory literature, the echelon and local base-stock levels for the most downstream stage are equal. We call it local because s_j^l depends on retailer j's local cost parameters. The warehouse orders the sum of these newsvendor solutions, that is, $\sum_{j=1}^{J} s_j^l(\boldsymbol{x}_j^{t+1})$. After the order arrives, the warehouse ships $s_j^l(\boldsymbol{x}_j^{t+1})$ to retailer j.

[3] The empirical in-sample cost is also named as empirical in-sample error or empirical error in the machine learning literature; see, for example, Bousquet and Elisseeff (2002). Here, we use "cost" instead of "error" for the context clarity.

[4] We call it estimation error because the cost gap is due to the deviation of the TA policy from the predictor (DDTA policy) by using the training data set S_N.

[5] We do not discuss the case where $0 < z_0 < \sum_{j=1}^{J}(\tilde{s}_j^e - \underline{m} \cdot r_j)$ because we shall show in Theorem 1 that the probability of $\tilde{z}_0$ that appears in this region converges to zero as $J \to \infty$.

[6] All computations were carried out on Python 3.7 with the solver SciPy to deal with optimization problems on a 1.6-GHz Intel Core i5 processor with 8.0 GB of random-access memory (RAM). The offline training for DDTA does not take much time. For example, for $N = 400$ and $J = 32$, the training time for quantile regression is within one minute. In Online Appendix B.6, we also test the cases with $N \in \{50, 100, 200, 800\}$.

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